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Project Proposal: Multi-threaded Online Quiz System

1. Introduction

In modern online education, thousands of students often take exams simultaneously. This requires a system that can handle multiple "Student" threads accessing a single "Question Bank" without crashing or corrupting data. This project simulates an online quiz platform where synchronization primitives (Mutexes and Semaphores) ensure that every student gets a fair, accurate, and thread-safe testing experience.

2. Methodology

The system will be developed in C++ using the pthread library. The logic is divided into three main synchronization areas:

Student Threads: Each student is a separate thread. They will "fetch" questions, "submit" answers, and "calculate" scores concurrently.

Question Bank (Thread-Safe Access): The questions are stored in a shared array. We will use a Mutex (or Read-Write Lock) to ensure that multiple students can read questions at once, but no data is corrupted during access.

Scoreboard (Mutual Exclusion): When a student finishes, they must update a global scoreboard. We will use a Mutex to lock the scoreboard so that only one student thread can write its final score at a time, preventing "lost updates."

Submission Queue (Semaphores): To simulate a server with limited capacity, we will use a Semaphore to limit how many students can submit their final exam to the "database" at the exact same time.

3. Additional Features (Any 5)

To meet the "Additional" requirements, the system will include:

Randomized Question Selection: Each student thread will receive questions in a different random order.

Timed Quizzes: Each thread will have a simulated "timer"; if they don't finish in time, the quiz auto-submits.

Analytics Dashboard: After all threads finish, the system will calculate the class average, highest score, and lowest score.

Logging and Tracing: The console will print live updates (e.g., "Student 5 started," "Student 2 submitted answer to Q3").

Configurable Student Count: The user can specify how many students (threads) are taking the quiz via the command line.